

LONESOME PINE, VA, USA

WMO: 724117

Lat: 36.988N

Lon: 82.530W

Elev: 2684

StdP: 13.33

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 63802

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.8	12.3	-2.2	5.4	19.2	1.2	6.5	19.3	24.9	43.7	22.5	42.5	6.1	300	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	86.3	69.4	83.9	69.0	81.9	68.2	73.7	80.4	72.6	79.1	71.5	77.8	5.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.3	132.3	76.3	71.5	128.7	75.6	70.0	122.1	74.4	39.2	80.3	38.2	79.1	37.1	78.9	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.1	17.7	15.7	DB	1.6	90.2	6.8	2.9	-3.3	92.3	-7.3	94.0	-11.1	95.7	-16.0	97.8	(4)
			WB	1.3	75.7	5.3	1.9	-2.5	77.1	-5.6	78.2	-8.5	79.3	-12.4	80.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	33.5	37.3	45.0	54.9	62.6	69.2	72.6	71.9	66.3	56.0	45.6	37.9	(6)
		DBStd	16.20	12.11	11.50	10.99	9.38	7.36	5.09	3.89	3.86	6.45	8.70	9.60	10.51	(7)
		HDD50	1826	525	377	227	59	7	0	0	1	43	194	393		(8)
		HDD65	4628	976	777	622	317	130	22	3	61	296	583	839		(9)
		CDD50	3466	14	20	72	205	399	576	701	680	489	229	62	19	(10)
		CDD65	793	0	0	2	14	57	148	239	217	99	16	1	0	(11)
		CDH74	4631	0	0	7	76	316	890	1418	1231	604	87	2	0	(12)
		CDH80	964	0	0	0	6	31	188	326	275	128	10	0	0	(13)
(14)	Wind	WSAvg	6.2	7.7	7.7	7.6	7.8	6.1	5.0	4.5	4.0	4.4	5.5	6.6	7.2	(14)
(15)	Precipitation	PrecAvg	48.4	4.5	4.1	4.9	4.1	4.4	4.3	5.1	4.0	3.0	2.3	3.2	4.5	(15)
(16)		PrecMax	59.2	6.8	9.9	10.2	9.0	7.5	6.9	8.0	7.1	7.0	4.4	6.1	9.5	(16)
(17)		PrecMin	36.1	2.2	1.7	2.0	1.9	1.5	2.5	1.9	1.7	0.8	0.3	1.1	1.7	(17)
(18)		PrecStd	6.8	1.3	1.9	2.0	1.9	1.7	1.3	1.6	1.3	1.7	1.0	1.3	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	62.8	67.9	74.5	80.7	82.4	88.2	89.5	88.4	87.6	81.1	71.7	64.0		(19)
(20)		MCWB	55.0	55.7	57.4	60.7	66.5	70.7	71.5	70.3	65.5	65.6	57.1	55.9		(20)
(21)		DB	58.8	62.6	69.8	75.4	80.9	84.2	86.2	85.6	83.5	75.5	66.0	60.9		(21)
(22)		MCWB	52.7	53.9	54.9	58.4	64.7	69.2	70.6	69.9	65.8	62.1	53.9	55.7		(22)
(23)		DB	55.0	57.3	64.3	72.7	78.2	81.8	83.4	82.4	81.1	72.5	63.3	56.6		(23)
(24)		MCWB	50.1	49.6	52.2	56.8	64.2	67.8	70.1	69.3	65.6	60.6	53.5	51.4		(24)
(25)		DB	52.0	54.0	61.2	69.9	74.7	79.4	81.4	81.1	77.3	69.7	60.5	54.1		(25)
(26)		MCWB	47.8	47.5	50.7	56.4	63.0	67.0	69.3	69.0	64.6	60.0	52.5	49.8		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	57.4	60.4	60.1	64.5	71.9	75.0	75.5	75.4	73.0	70.3	61.3	60.7		(27)
(28)		MCDB	60.0	63.2	68.9	73.5	78.4	82.4	82.8	81.6	79.0	75.8	64.1	61.9		(28)
(29)		WB	54.5	55.9	57.5	62.4	68.8	72.5	73.7	73.3	71.1	67.2	58.9	56.7		(29)
(30)		MCDB	57.7	60.2	65.3	71.2	75.7	79.8	80.5	79.8	77.0	73.1	63.3	59.4		(30)
(31)		WB	51.6	52.2	55.2	60.5	67.0	70.7	72.6	72.2	69.3	63.9	56.1	53.9		(31)
(32)		MCDB	54.9	55.5	61.7	68.1	74.3	77.6	79.1	78.5	74.5	68.0	61.5	55.8		(32)
(33)		WB	47.7	48.5	52.9	58.4	65.0	69.3	71.6	70.9	67.3	61.4	53.6	50.1		(33)
(34)		MCDB	50.7	52.5	59.4	66.1	71.3	76.1	77.8	77.0	73.6	66.6	57.9	53.4		(34)
(35)	Mean Daily Temperature Range	MDBR	15.4	15.7	17.8	19.1	18.3	18.0	16.8	17.0	17.8	17.9	16.8	14.7		(35)
(36)		5% DB	16.3	18.8	20.7	21.4	20.3	20.4	19.5	19.1	21.0	20.6	18.8	15.8		(36)
(37)		MCWBR	12.3	13.1	11.7	10.5	9.4	9.1	8.0	8.1	8.2	9.8	11.3	11.9		(37)
(38)		5% WB	14.3	15.8	17.3	17.6	17.6	17.6	16.2	16.0	16.2	16.5	15.3	13.8		(38)
(39)		MCWBR	12.5	13.9	11.7	10.8	9.6	9.5	8.1	8.2	8.8	10.3	11.6	12.9		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.285	0.296	0.318	0.355	0.392	0.417	0.446	0.450	0.364	0.334	0.302	0.282		(40)
(41)		taud	2.553	2.515	2.491	2.391	2.283	2.242	2.193	2.161	2.396	2.486	2.554	2.581		(41)
(42)		Ebn at Noon	292	300	301	293	283	274	265	261	282	282	282	285		(42)
(43)		Edh at Noon	24	28	32	37	42	44	46	46	35	29	24	22		(43)
(44)	All-Sky Solar Radiation	RadAvg	661	848	1226	1591	1776	1936	1805	1708	1448	1105	803	582		(44)
(45)		RadStd	59	116	105	115	139	115	110	82	127	137	90	61		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+1.12	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+247	+142

Nomenclature: See separate page